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Applied Calculus IV

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Systems of Linear First-Order Differential Equations

➤ Solutions

- ✓ when eigenvalues are real numbers
- ✓ When eigenvalues are complex number

Recall (Definition of Complex Conjugate)

Every complex number $z = a + bi$ has a "twin" called the **complex conjugate**, denoted as $\bar{z}$

– Finding the conjugate:

1. Keep the **real** part exactly the same
2. Flip the sign of the **imaginary** part.

$$\bar{z} = a - bi$$

Example: $z = 3 + 2i \rightarrow \bar{z} = 3 - 2i$

Theorem (Solutions corresponding to a complex)

Let $\mathbf{A}$ be the coefficient matrix (with real entries) in the homogeneous system

$$\mathbf{X}' = \mathbf{A}\mathbf{X},$$

and let $\mathbf{K}_1$ be an eigenvector corresponding to the complex eigenvalue $\lambda_1 = \alpha + i\beta$,

α and β are real. Then

$$\mathbf{K}_1 e^{\lambda_1 t} \text{ and } \overline{\mathbf{K}_1} e^{\overline{\lambda_1} t}$$

are solutions of the system.

Corollary (optional): For $\lambda_1 = \alpha + i\beta$ and $\mathbf{K}_1$, solution vectors $\mathbf{X}_1$ and $\mathbf{X}_2$ can be shown as

$$\mathbf{X}_1 = [\mathbf{B}_1 \cos \beta t - \mathbf{B}_2 \sin \beta t] e^{\alpha t}$$

$$\mathbf{X}_2 = [\mathbf{B}_2 \cos \beta t + \mathbf{B}_1 \sin \beta t] e^{\alpha t}$$

$\mathbf{B}_1$ is the real part of the eigenvector $\mathbf{K}_1$

$\mathbf{B}_2$ is the imaginary part of the eigenvector $\mathbf{K}_1$

$$\mathbf{B}_1 = \text{Re}(\mathbf{K}_1)$$

$$\mathbf{B}_2 = \text{Im}(\mathbf{K}_1)$$

Example: Solve the given system

$$\begin{cases} x' = 6x - y \\ y' = 5x + 4y \end{cases}$$

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 6 & -1 \\ 5 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$



Change it to the matrix form

Solution:

Step 1. Find eigenvalues λ_i

$$\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} 6 - \lambda & -1 \\ 5 & 4 - \lambda \end{vmatrix} = \lambda^2 - 10\lambda + 29 = 0.$$

 $\lambda_1 = 5 + 2i, \lambda_2 = 5 - 2i$

Step 2. Find eigenvectors $\mathbf{K}_1$

$$\lambda_1 = 5 + 2i \quad \begin{pmatrix} 1 - 2i & -1 \\ 5 & -1 - 2i \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \mathbf{0} \quad \longrightarrow \quad (1 - 2i)k_1 = k_2$$

$$\Rightarrow \quad \mathbf{K} = \begin{pmatrix} 1 \\ 1 - 2i \end{pmatrix}$$

Step 3. Construct the general solution formula:

$$\mathbf{X} = c_1 \mathbf{X}_1 + c_2 \mathbf{X}_2$$

$$\mathbf{X}(t) = c_1 \mathbf{K}_1 e^{\lambda_1 t} + c_2 \overline{\mathbf{K}_1} e^{\overline{\lambda_1} t}$$


$$\mathbf{X}(t) = c_1 \begin{pmatrix} 1 \\ 1 - 2i \end{pmatrix} e^{(5+2i)t} + c_2 \begin{pmatrix} 1 \\ 1 + 2i \end{pmatrix} e^{(5-2i)t}$$

Or **(optional)** construct the general solution as: $\mathbf{X} = c_1\mathbf{X}_1 + c_2\mathbf{X}_2$

where

$$\mathbf{X}_1 = [\mathbf{B}_1 \cos \beta t - \mathbf{B}_2 \sin \beta t] e^{\alpha t}$$

$$\mathbf{X}_2 = [\mathbf{B}_2 \cos \beta t + \mathbf{B}_1 \sin \beta t] e^{\alpha t}$$

$$\begin{aligned}\mathbf{K} &= \begin{pmatrix} 1 \\ 1 - 2i \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + i \begin{pmatrix} 0 \\ -2 \end{pmatrix} \\ &= \mathbf{B}_1 + i\mathbf{B}_2\end{aligned}$$

$$\mathbf{X}_1 = e^{5t} \begin{pmatrix} \cos 2t \\ \cos 2t + 2 \sin 2t \end{pmatrix}, \quad \mathbf{X}_2 = e^{5t} \begin{pmatrix} \sin 2t \\ -2 \cos 2t + \sin 2t \end{pmatrix}$$

 $\mathbf{X}(t) = C_1 e^{5t} \begin{pmatrix} \cos 2t \\ \cos 2t + 2 \sin 2t \end{pmatrix} + C_2 e^{5t} \begin{pmatrix} \sin 2t \\ -2 \cos 2t + \sin 2t \end{pmatrix}$

Example: Solve the initial value problem

$$\mathbf{X}' = \begin{pmatrix} 2 & 8 \\ -1 & -2 \end{pmatrix} \mathbf{X},$$

Solution:

Step 1. Find eigenvalues λ_i

$$\det \begin{pmatrix} 2 - \lambda & 8 \\ -1 & -2 - \lambda \end{pmatrix} = (2 - \lambda)(-2 - \lambda) + 8 = \lambda^2 + 4 = 0$$



$$\lambda = 2i, -2i$$

Step 2. Find eigenvectors $\mathbf{K}_1$

$$\lambda_1 = 2i \longrightarrow (\mathbf{A} - 2i\mathbf{I})\mathbf{K} = \mathbf{0}$$

$$\begin{pmatrix} 2 - 2i & 8 \\ -1 & -2 - 2i \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \mathbf{0} \longrightarrow -k_1 - (2 + 2i)k_2 = 0$$

$$\longrightarrow \mathbf{K} = \begin{pmatrix} -(2 + 2i) \\ 1 \end{pmatrix}$$

Step 3. Construct the general solution formula:

$$\mathbf{X} = c_1 \mathbf{X}_1 + c_2 \mathbf{X}_2$$

$$\mathbf{X}(t) = c_1 \mathbf{K}_1 e^{\lambda_1 t} + c_2 \overline{\mathbf{K}_1} e^{\overline{\lambda_1} t}$$


$$\mathbf{X}(t) = c_1 \begin{pmatrix} -(2 + 2i) \\ 1 \end{pmatrix} e^{2it} + c_2 \begin{pmatrix} -(2 - 2i) \\ 1 \end{pmatrix} e^{-2it}$$

Or **(optional)** construct the general solution as: $\mathbf{X} = c_1 \mathbf{X}_1 + c_2 \mathbf{X}_2$

where

$$\mathbf{X}_1 = [\mathbf{B}_1 \cos \beta t - \mathbf{B}_2 \sin \beta t] e^{\alpha t}$$

$$\mathbf{X}_2 = [\mathbf{B}_2 \cos \beta t + \mathbf{B}_1 \sin \beta t] e^{\alpha t}$$

$$\begin{aligned} \mathbf{K} &= \begin{pmatrix} -(2 + 2i) \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ 1 \end{pmatrix} + i \begin{pmatrix} -2 \\ 0 \end{pmatrix} \\ &= \mathbf{B}_1 + i\mathbf{B}_2 \end{aligned}$$

$$\mathbf{X}_1 = \begin{pmatrix} -2 \cos 2t + 2 \sin 2t \\ \cos 2t \end{pmatrix}, \quad \mathbf{X}_2 = \begin{pmatrix} -2 \cos 2t - 2 \sin 2t \\ \sin 2t \end{pmatrix}$$

➡
$$\mathbf{X}(t) = C_1 \begin{pmatrix} -2 \cos 2t + 2 \sin 2t \\ \cos 2t \end{pmatrix} + C_2 \begin{pmatrix} -2 \cos 2t - 2 \sin 2t \\ \sin 2t \end{pmatrix}$$

Example: Solve the given system of ODEs

$$\begin{cases} x' = z \\ y' = -z \\ z' = y \end{cases}$$

$$\mathbf{X}' = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \mathbf{X}$$



Change it to the matrix form

Solution:

Step 1. Find eigenvalues λ_i $\det(A - \lambda I) = \begin{vmatrix} -\lambda & 0 & 1 \\ 0 & -\lambda & -1 \\ 0 & 1 & -\lambda \end{vmatrix} = -\lambda^3 - \lambda$

 $\lambda_1 = 0, \quad \lambda_2 = i, \quad \lambda_3 = -i$

Step 2. Find eigenvectors $\mathbf{K}_1$ and $\mathbf{K}_2$

$$\lambda_1 = 0, \quad \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \\ k_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \longrightarrow \mathbf{K}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\lambda_2 = i, \quad \begin{pmatrix} -i & 0 & 1 \\ 0 & -i & -1 \\ 0 & 1 & -i \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \\ k_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \longrightarrow \mathbf{K}_2 = \begin{pmatrix} -i \\ i \\ 1 \end{pmatrix}$$

Step 3. Construct the general solution formula:

$$\mathbf{X} = c_1 \mathbf{X}_1 + c_2 \mathbf{X}_2 + c_3 \mathbf{X}_3$$

$$\mathbf{X} = c_1 \mathbf{K}_1 e^{\lambda_1 t} + c_2 \mathbf{K}_2 e^{\lambda_2 t} + c_3 \overline{\mathbf{K}_2} e^{\overline{\lambda_2} t}$$


$$\mathbf{X} = c_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} e^0 + c_2 \begin{pmatrix} -i \\ i \\ 1 \end{pmatrix} e^{it} + c_3 \begin{pmatrix} i \\ -i \\ 1 \end{pmatrix} e^{-it}$$

Exercises 8.2

(Part Three)

In Problems 35–46 find the general solution of the given system.

$$35. \begin{cases} x' = 6x - y \\ y' = 5x + 2y \end{cases}$$

$$39. \mathbf{X}' = \begin{pmatrix} 4 & -5 \\ 5 & -4 \end{pmatrix} \mathbf{X}$$

$$43. \mathbf{X}' = \begin{pmatrix} 1 & -1 & 2 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} \mathbf{X}$$

$$44. \mathbf{X}' = \begin{pmatrix} 4 & 0 & 1 \\ 0 & 6 & 0 \\ -4 & 0 & 4 \end{pmatrix} \mathbf{X}$$

$$36. \begin{cases} x' = x + y \\ y' = -2x - y \end{cases}$$

$$40. \mathbf{X}' = \begin{pmatrix} 1 & -8 \\ 1 & -3 \end{pmatrix} \mathbf{X}$$

$$45. \mathbf{X}' = \begin{pmatrix} 2 & 5 & 1 \\ -5 & -6 & 4 \\ 0 & 0 & 2 \end{pmatrix} \mathbf{X}$$

$$46. \mathbf{X}' = \begin{pmatrix} 2 & 4 & 4 \\ -1 & -2 & 0 \\ -1 & 0 & -2 \end{pmatrix} \mathbf{X}$$

$$37. \begin{cases} x' = 5x + y \\ y' = -2x + 3y \end{cases}$$

$$41. \begin{cases} x' = z \\ y' = -z \\ z' = y \end{cases}$$

$$38. \begin{cases} x' = 4x + 5y \\ y' = -2x + 6y \end{cases}$$

$$42. \begin{cases} x' = 2x + y + 2z \\ y' = 3x + 6z \\ z' = -4x - 3z \end{cases}$$

In Problems 47 and 48 solve the given initial-value problem.

$$47. \mathbf{X}' = \begin{pmatrix} 1 & -12 & -14 \\ 1 & 2 & -3 \\ 1 & 1 & -2 \end{pmatrix} \mathbf{X}, \quad \mathbf{X}(0) = \begin{pmatrix} 4 \\ 6 \\ -7 \end{pmatrix}$$

$$48. \mathbf{X}' = \begin{pmatrix} 6 & -1 \\ 5 & 4 \end{pmatrix} \mathbf{X}, \quad \mathbf{X}(0) = \begin{pmatrix} -2 \\ 8 \end{pmatrix}$$

Systems of Linear First-Order Differential Equations

➤ Nonhomogeneous Linear System

- Undetermined Coefficients
- Variation of Parameters

Undetermined Coefficients

Example: Solve the system

$$\mathbf{X}' = \begin{pmatrix} -1 & 2 \\ -1 & 1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} -8 \\ 3 \end{pmatrix}$$

Step 1. Find $\mathbf{X}_c$

$$\det(\mathbf{A} - \lambda \mathbf{I}) = \begin{vmatrix} -1 - \lambda & 2 \\ -1 & 1 - \lambda \end{vmatrix} = \lambda^2 + 1 = 0, \quad \longrightarrow \quad \lambda = \pm i$$

$$\begin{pmatrix} -1 - i & 2 \\ -1 & 1 - i \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \mathbf{0} \quad \longrightarrow \quad k_1 = (1 - i)k_2$$

$$\longrightarrow \quad \mathbf{K} = \begin{pmatrix} 1 - i \\ 1 \end{pmatrix}$$

$$\mathbf{X}_c = c_1 \mathbf{X}_1 + c_2 \mathbf{X}_2 \quad \longrightarrow \quad \mathbf{X}_c(t) = c_1 \mathbf{K} e^{\lambda_1 t} + c_2 \overline{\mathbf{K}} e^{\overline{\lambda}_1 t}$$

$$\longrightarrow \quad \mathbf{X}_c(t) = c_1 \begin{pmatrix} 1 - i \\ 1 \end{pmatrix} e^{it} + c_2 \begin{pmatrix} 1 + i \\ 1 \end{pmatrix} e^{-it}$$

Step 2. Find $\mathbf{X}_p$

Define $\mathbf{X}_p = \begin{pmatrix} a \\ b \end{pmatrix}$ Replace it in the system $\mathbf{0} = \begin{pmatrix} -1 & 2 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} -8 \\ 3 \end{pmatrix}$

$$\begin{pmatrix} -1 & 2 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 8 \\ -3 \end{pmatrix}$$

Solve the system $\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 14 \\ 11 \end{pmatrix}$

Step 3. Find $\mathbf{X}$

$$\mathbf{X}(t) = \mathbf{X}_c(t) + \mathbf{X}_p = c_1 \begin{pmatrix} 1 - i \\ 1 \end{pmatrix} e^{it} + c_2 \begin{pmatrix} 1 + i \\ 1 \end{pmatrix} e^{-it} + \begin{pmatrix} 14 \\ 11 \end{pmatrix}$$

Example: Solve the system

$$\mathbf{X}' = \begin{pmatrix} 6 & 1 \\ 4 & 3 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 6t \\ -10t + 4 \end{pmatrix}$$



$$\mathbf{F}(t) = \begin{pmatrix} 6 \\ -10 \end{pmatrix} t + \begin{pmatrix} 0 \\ 4 \end{pmatrix},$$

Solution:

$$\mathbf{A} = \begin{pmatrix} 6 & 1 \\ 4 & 3 \end{pmatrix}$$

Step 1. Find $\mathbf{X}_c$

$$\det(\mathbf{A} - \lambda \mathbf{I}) = (6 - \lambda)(3 - \lambda) - 4 = \lambda^2 - 9\lambda + 14 = 0 \longrightarrow \lambda_1 = 2, \lambda_2 = 7$$

$$\lambda = 2: \begin{pmatrix} 4 & 1 \\ 4 & 1 \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \mathbf{0} \longrightarrow 4k_1 + k_2 = 0 \longrightarrow \mathbf{K}_1 = \begin{pmatrix} 1 \\ -4 \end{pmatrix}$$

$$\lambda = 7: \begin{pmatrix} -1 & 1 \\ 4 & -4 \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \mathbf{0} \longrightarrow -k_1 + k_2 = 0 \longrightarrow \mathbf{K}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\longrightarrow \mathbf{X}_c(t) = C_1 e^{2t} \begin{pmatrix} 1 \\ -4 \end{pmatrix} + C_2 e^{7t} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Step 2. Find $\mathbf{X}_p$

Define $\mathbf{X}_p(t) = \begin{pmatrix} at + b \\ ct + d \end{pmatrix} \longrightarrow \mathbf{X}'_p(t) = \begin{pmatrix} a \\ c \end{pmatrix}$

Replace them
in the system

$$\begin{pmatrix} a \\ c \end{pmatrix} = \begin{pmatrix} 6 & 1 \\ 4 & 3 \end{pmatrix} \begin{pmatrix} at + b \\ ct + d \end{pmatrix} + \begin{pmatrix} 6t \\ -10t + 4 \end{pmatrix}$$
$$= \begin{pmatrix} 6at + 6b + ct + d + 6t \\ 4at + 4b + 3ct + 3d - 10t + 4 \end{pmatrix} = \begin{pmatrix} (6a + c + 6)t + (6b + d) \\ (4a + 3c - 10)t + (4b + 3d + 4) \end{pmatrix}$$

$$\begin{cases} 6a + c = -6 \\ 4a + 3c = 10 \end{cases} \longrightarrow a = -2, c = 6$$

$$\begin{cases} 6b + d = -2 \\ 4b + 3d = 6 - 4 = 2 \end{cases} \longrightarrow \begin{matrix} b = -\frac{4}{7} \\ d = \frac{10}{7} \end{matrix}$$

$\longrightarrow \mathbf{X}_p(t) = \begin{pmatrix} -2t - \frac{4}{7} \\ 6t + \frac{10}{7} \end{pmatrix}$

Step 3. Find the general solution $\mathbf{X}$

$$\mathbf{X}(t) = \mathbf{X}_c(t) + \mathbf{X}_p(t) = C_1 e^{2t} \begin{pmatrix} 1 \\ -4 \end{pmatrix} + C_2 e^{7t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} -2t - \frac{4}{7} \\ 6t + \frac{10}{7} \end{pmatrix}$$

Or

$$\mathbf{X}(t) = \underbrace{\begin{pmatrix} e^{2t} & e^{7t} \\ -4e^{2t} & e^{7t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}}_{\mathbf{X}_c(t)} + \underbrace{\begin{pmatrix} -2t - \frac{4}{7} \\ 6t + \frac{10}{7} \end{pmatrix}}_{\mathbf{X}_p(t)}$$

Example: Determine the form of a particular solution vector $\mathbf{X}_p$ for the system

$$\begin{cases} x' = 5x + 3y - 2e^{-t} + 1 \\ y' = -x + y + e^{-t} - 5t + 7 \end{cases}$$

Change it to the matrix form

$$\mathbf{X}' = \begin{pmatrix} 5 & 3 \\ -1 & 1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} -2e^{-t} + 1 \\ e^{-t} - 5t + 7 \end{pmatrix}$$



$$\mathbf{F}(t) = \begin{pmatrix} -2 \\ 1 \end{pmatrix} e^{-t} + \begin{pmatrix} 0 \\ -5 \end{pmatrix} t + \begin{pmatrix} 1 \\ 7 \end{pmatrix}$$

$$\Rightarrow \mathbf{X}_p = \begin{pmatrix} a_3 \\ b_3 \end{pmatrix} e^{-t} + \begin{pmatrix} a_2 \\ b_2 \end{pmatrix} t + \begin{pmatrix} a_1 \\ b_1 \end{pmatrix}$$

Variation of Parameters

Recall (Properties a fundamental matrix)

1. A fundamental matrix $\Phi(t)$ is nonsingular.
2. If $\Phi(t)$ is a fundamental matrix of the system $\mathbf{X}' = \mathbf{A}\mathbf{X}$, then

$$\Phi'(t) = \mathbf{A}\Phi(t).$$

Define

$$\mathbf{U}(t) = \begin{pmatrix} u_1(t) \\ u_2(t) \\ \vdots \\ u_n(t) \end{pmatrix}$$

and define: $\Phi(t)$ is a fundamental matrix of the system $\mathbf{X}' = \mathbf{A}\mathbf{X}$,

Assume

$$\mathbf{X}_p = \Phi(t)\mathbf{U}(t) \quad \text{Be a particular solution of } \mathbf{X}' = \mathbf{A}\mathbf{X} + \mathbf{F}(t)$$



$$\mathbf{X}'_p = \Phi(t)\mathbf{U}'(t) + \Phi'(t)\mathbf{U}(t).$$



$$\Phi(t)\mathbf{U}'(t) + \boxed{\Phi'(t)\mathbf{U}(t)} = \mathbf{A}\Phi(t)\mathbf{U}(t) + \mathbf{F}(t)$$



$$\Phi(t)\mathbf{U}'(t) + \mathbf{A}\Phi(t)\mathbf{U}(t) = \mathbf{A}\Phi(t)\mathbf{U}(t) + \mathbf{F}(t)$$

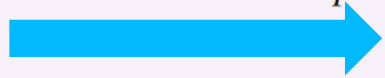
$$\Phi(t)\mathbf{U}'(t) = \mathbf{F}(t).$$

$$\rightarrow \mathbf{U}'(t) = \mathbf{\Phi}^{-1}(t)\mathbf{F}(t)$$

$$\rightarrow \mathbf{U}(t) = \int \mathbf{\Phi}^{-1}(t)\mathbf{F}(t) dt$$

$$\rightarrow \mathbf{X}_p = \mathbf{\Phi}(t) \int \mathbf{\Phi}^{-1}(t)\mathbf{F}(t) dt$$

$$\mathbf{X} = \mathbf{X}_c + \mathbf{X}_p$$



$$\mathbf{X} = \mathbf{\Phi}(t)\mathbf{C} + \mathbf{\Phi}(t) \int \mathbf{\Phi}^{-1}(t)\mathbf{F}(t) dt$$

is the general solution of $\mathbf{X}' = \mathbf{A}\mathbf{X} + \mathbf{F}(t)$

where $\mathbf{\Phi}(t)$ is a fundamental matrix of the system $\mathbf{X}' = \mathbf{A}\mathbf{X}$,

Example: Solve the system

$$\mathbf{X}' = \begin{pmatrix} -3 & 1 \\ 2 & -4 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 3t \\ e^{-t} \end{pmatrix}$$

Solution:

$$\mathbf{A} = \begin{pmatrix} -3 & 1 \\ 2 & -4 \end{pmatrix}$$

Step 1. Find $\mathbf{X}_c$

$$\det(\mathbf{A} - \lambda \mathbf{I}) = (-3 - \lambda)(-4 - \lambda) - 2 = \lambda^2 + 7\lambda + 10 = 0 \quad \longrightarrow \quad \lambda = -2, -5$$

$$\lambda = -2, \quad \begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \mathbf{0} \quad \longrightarrow \quad -k_1 + k_2 = 0 \quad \longrightarrow \quad \mathbf{K}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\lambda = -5 \quad \begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \mathbf{0} \quad \longrightarrow \quad 2k_1 + k_2 = 0 \quad \longrightarrow \quad \mathbf{K}_2 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

$$\longrightarrow \quad \Phi(t) = \begin{pmatrix} e^{-2t} & e^{-5t} \\ e^{-2t} & -2e^{-5t} \end{pmatrix}$$

$$\Phi(t) = \begin{pmatrix} e^{-2t} & e^{-5t} \\ e^{-2t} & -2e^{-5t} \end{pmatrix}$$

$$\Phi^{-1}(t) = -\frac{1}{3}e^{7t} \begin{pmatrix} -2e^{-5t} & -e^{-5t} \\ -e^{-2t} & e^{-2t} \end{pmatrix} = \begin{pmatrix} \frac{2}{3}e^{2t} & \frac{1}{3}e^{2t} \\ \frac{1}{3}e^{5t} & -\frac{1}{3}e^{5t} \end{pmatrix}$$

$$\Phi^{-1}(t)\mathbf{F}(t) = \begin{pmatrix} \frac{2}{3}e^{2t} & \frac{1}{3}e^{2t} \\ \frac{1}{3}e^{5t} & -\frac{1}{3}e^{5t} \end{pmatrix} \begin{pmatrix} 3t \\ e^{-t} \end{pmatrix} = \begin{pmatrix} 2te^{2t} + \frac{1}{3}e^t \\ te^{5t} - \frac{1}{3}e^{4t} \end{pmatrix}$$

$$\int \Phi^{-1}(t)\mathbf{F}(t)dt = \begin{pmatrix} te^{2t} - \frac{1}{2}e^{2t} + \frac{1}{3}e^t \\ \frac{1}{5}te^{5t} - \frac{1}{25}e^{5t} - \frac{1}{12}e^{4t} \end{pmatrix}$$

$$\Rightarrow \mathbf{X}_p(t) = \begin{pmatrix} e^{-2t} & e^{-5t} \\ e^{-2t} & -2e^{-5t} \end{pmatrix} \begin{pmatrix} te^{2t} - \frac{1}{2}e^{2t} + \frac{1}{3}e^t \\ \frac{1}{5}te^{5t} - \frac{1}{25}e^{5t} - \frac{1}{12}e^{4t} \end{pmatrix} = \begin{pmatrix} \frac{6}{5}t - \frac{27}{50} + \frac{1}{4}e^{-t} \\ \frac{3}{5}t - \frac{21}{50} + \frac{1}{2}e^{-t} \end{pmatrix}$$

$$\mathbf{X} = \underbrace{\begin{pmatrix} e^{-2t} & e^{-5t} \\ e^{-2t} & -2e^{-5t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}}_{\text{Homogeneous Solution}} + \underbrace{\begin{pmatrix} \frac{6}{5}t - \frac{27}{50} + \frac{1}{4}e^{-t} \\ \frac{3}{5}t - \frac{21}{50} + \frac{1}{2}e^{-t} \end{pmatrix}}_{\text{Particular Solution}}$$

Or
$$= c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-2t} + c_2 \begin{pmatrix} 1 \\ -2 \end{pmatrix} e^{-5t} + \begin{pmatrix} \frac{6}{5} \\ \frac{3}{5} \end{pmatrix} t - \begin{pmatrix} \frac{27}{50} \\ \frac{21}{50} \end{pmatrix} + \begin{pmatrix} \frac{1}{4} \\ \frac{1}{2} \end{pmatrix} e^{-t}$$

Solution of the Systems of Linear First-Order ODEs

Homogenous

$$\mathbf{X}' = \mathbf{A}\mathbf{X},$$

$$\mathbf{X}(t) = \Phi(t)\mathbf{C}$$

Nonhomogenous

$$\mathbf{X}' = \mathbf{A}\mathbf{X} + \mathbf{F}(t),$$

$$\mathbf{X} = \Phi(t)\mathbf{C} + \Phi(t) \int \Phi^{-1}(t)\mathbf{F}(t) dt$$

Exercises 8.3

In Problems 1–8, solve the given nonhomogeneous system.

1. $\frac{dx}{dt} = 2x + 3y - 7$

$$\frac{dy}{dt} = -x - 2y + 5$$

2. $\frac{dx}{dt} = 5x + 9y + 2$

$$\frac{dy}{dt} = -x + 11y + 6$$

3. $\mathbf{X}' = \begin{pmatrix} 1 & 3 \\ 3 & 1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} -2t^2 \\ t + 5 \end{pmatrix}$

4. $\mathbf{X}' = \begin{pmatrix} 1 & -4 \\ 4 & 1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 4t + 9e^{6t} \\ -t + e^{6t} \end{pmatrix}$

5. $\mathbf{X}' = \begin{pmatrix} 4 & \frac{1}{3} \\ 9 & 6 \end{pmatrix} \mathbf{X} + \begin{pmatrix} -3 \\ 10 \end{pmatrix} e^t$

6. $\mathbf{X}' = \begin{pmatrix} -1 & 5 \\ -1 & 1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} \sin t \\ -2 \cos t \end{pmatrix}$

7. $\mathbf{X}' = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 3 \\ 0 & 0 & 5 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} e^{4t}$

8. $\mathbf{X}' = \begin{pmatrix} 0 & 0 & 5 \\ 0 & 5 & 0 \\ 5 & 0 & 0 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 5 \\ -10 \\ 40 \end{pmatrix}$

The method of
undetermined coefficients
can be applied

In Problems 9 and 10, solve the given initial-value problem.

$$\mathbf{9.} \quad \mathbf{X}' = \begin{pmatrix} -1 & -2 \\ 3 & 4 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 3 \\ 3 \end{pmatrix}, \quad \mathbf{X}(0) = \begin{pmatrix} -4 \\ 5 \end{pmatrix}$$

$$\mathbf{10.} \quad \mathbf{X}' = \begin{pmatrix} 1 & -1 \\ 1 & 3 \end{pmatrix} \mathbf{X} + \begin{pmatrix} t \\ t+1 \end{pmatrix}, \quad \mathbf{X}(0) = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

In Problems 13–32 use variation of parameters to solve the given nonhomogeneous system

$$13. \frac{dx}{dt} = 3x - 3y + 4$$

$$\frac{dy}{dt} = 2x - 2y - 1$$

$$14. \frac{dx}{dt} = 2x - y$$

$$\frac{dy}{dt} = 3x - 2y + 4t$$

$$15. \mathbf{X}' = \begin{pmatrix} 3 & -5 \\ \frac{3}{4} & -1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{t/2}$$

$$16. \mathbf{X}' = \begin{pmatrix} 2 & -1 \\ 4 & 2 \end{pmatrix} \mathbf{X} + \begin{pmatrix} \sin 2t \\ 2 \cos 2t \end{pmatrix} e^{2t}$$

$$17. \mathbf{X}' = \begin{pmatrix} 0 & 2 \\ -1 & 3 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^t$$

$$18. \mathbf{X}' = \begin{pmatrix} 0 & 2 \\ -1 & 3 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 2 \\ e^{-3t} \end{pmatrix}$$

$$19. \mathbf{X}' = \begin{pmatrix} 1 & 8 \\ 1 & -1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 12 \\ 12 \end{pmatrix} t$$

$$20. \mathbf{X}' = \begin{pmatrix} 1 & 8 \\ 1 & -1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} e^{-t} \\ te^t \end{pmatrix}$$

$$21. \mathbf{X}' = \begin{pmatrix} 3 & 2 \\ -2 & -1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 2e^{-t} \\ e^{-t} \end{pmatrix}$$

$$22. \mathbf{X}' = \begin{pmatrix} 3 & 2 \\ -2 & -1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$23. \mathbf{X}' = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \mathbf{X} + \begin{pmatrix} \sec t \\ 0 \end{pmatrix}$$

$$24. \mathbf{X}' = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 3 \\ 3 \end{pmatrix} e^t$$

$$25. \mathbf{X}' = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} e^t$$

$$27. \mathbf{X}' = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 0 \\ \sec t \tan t \end{pmatrix}$$

$$28. \mathbf{X}' = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 1 \\ \cot t \end{pmatrix}$$

$$29. \mathbf{X}' = \begin{pmatrix} 1 & 2 \\ -\frac{1}{2} & 1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} \csc t \\ \sec t \end{pmatrix} e^t$$

$$30. \mathbf{X}' = \begin{pmatrix} 1 & -2 \\ 1 & -1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} \tan t \\ 1 \end{pmatrix}$$

$$31. \mathbf{X}' = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix} \mathbf{X} + \begin{pmatrix} e^t \\ e^{2t} \\ te^{3t} \end{pmatrix}$$

$$32. \mathbf{X}' = \begin{pmatrix} 3 & -1 & -1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 0 \\ t \\ 2e^t \end{pmatrix}$$

Systems of Linear First-Order Differential Equations

➤ Matrix Exponential

Recall (Series of e^{at})

$$\begin{aligned} e^{at} &= 1 + at + \frac{(at)^2}{2!} + \frac{(at)^3}{3!} + \dots + \frac{(at)^k}{k!} + \dots \\ &= 1 + at + a^2 \frac{t^2}{2!} + a^3 \frac{t^3}{3!} + \dots + a^k \frac{t^k}{k!} + \dots \\ &= \sum_{k=0}^{\infty} a^k \frac{t^k}{k!} \end{aligned}$$

Theorem (Matrix Exponential)

For any $n \times n$ matrix $\mathbf{A}$,

$$e^{\mathbf{A}t} = \mathbf{I} + \mathbf{A}t + \mathbf{A}^2 \frac{t^2}{2!} + \cdots + \mathbf{A}^k \frac{t^k}{k!} + \cdots = \sum_{k=0}^{\infty} \mathbf{A}^k \frac{t^k}{k!}$$

Example: Find e^{At} For the matrix

$$\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$$

Solution

$$\mathbf{A}^2 = \begin{pmatrix} 2^2 & 0 \\ 0 & 3^2 \end{pmatrix}, \mathbf{A}^3 = \begin{pmatrix} 2^3 & 0 \\ 0 & 3^3 \end{pmatrix}, \mathbf{A}^4 = \begin{pmatrix} 2^4 & 0 \\ 0 & 3^4 \end{pmatrix}, \dots, \mathbf{A}^n = \begin{pmatrix} 2^n & 0 \\ 0 & 3^n \end{pmatrix}, \dots,$$

$$\begin{aligned} e^{At} &= \mathbf{I} + \mathbf{A}t + \mathbf{A}^2 \frac{t^2}{2!} + \dots \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} t + \begin{pmatrix} 2^2 & 0 \\ 0 & 3^2 \end{pmatrix} \frac{t^2}{2!} + \dots + \begin{pmatrix} 2^n & 0 \\ 0 & 3^n \end{pmatrix} \frac{t^n}{n!} + \dots \\ &= \begin{pmatrix} 1 + 2t + 2^2 \frac{t^2}{2!} + \dots & 0 \\ 0 & 1 + 3t + 3^2 \frac{t^2}{2!} + \dots \end{pmatrix} = \begin{pmatrix} e^{2t} & 0 \\ 0 & e^{3t} \end{pmatrix} \end{aligned}$$

Remark (diagonal matrix)

$$\text{If } \mathbf{A} = \begin{pmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{pmatrix}, \text{ then } e^{\mathbf{A}t} = \begin{pmatrix} e^{a_{11}t} & 0 & \cdots & 0 \\ 0 & e^{a_{22}t} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & e^{a_{nn}t} \end{pmatrix}$$

Remark $(e^{\mathbf{A}t})'$

$$\frac{d}{dt}e^{\mathbf{A}t} = \mathbf{A}e^{\mathbf{A}t}$$

Because

$$\begin{aligned}\frac{d}{dt}e^{\mathbf{A}t} &= \frac{d}{dt} \left[\mathbf{I} + \mathbf{A}t + \mathbf{A}^2 \frac{t^2}{2!} + \dots + \mathbf{A}^k \frac{t^k}{k!} + \dots \right] = \mathbf{A} + \mathbf{A}^2 t + \frac{1}{2!} \mathbf{A}^3 t^2 + \dots \\ &= \mathbf{A} \left[\mathbf{I} + \mathbf{A}t + \mathbf{A}^2 \frac{t^2}{2!} + \dots \right] = \mathbf{A}e^{\mathbf{A}t}.\end{aligned}$$

Remark

$$\mathbf{X} = e^{\mathbf{A}t} \mathbf{C}$$

is the general solution of system $\mathbf{X}' = \mathbf{A}\mathbf{X}$,

- $\mathbf{A}$ is an $n \times n$ matrix of constant coefficients,
- $\mathbf{X}$ is an $n \times 1$ matrix (the solution vector),
- $\mathbf{C}$ is an $n \times 1$ matrix (with arbitrary constant entries),

Because
$$\mathbf{X}' = \frac{d}{dt} e^{\mathbf{A}t} \mathbf{C} = \mathbf{A} e^{\mathbf{A}t} \mathbf{C} = \mathbf{A} (e^{\mathbf{A}t} \mathbf{C}) = \mathbf{A} \mathbf{X}.$$

Remark

$$\mathbf{X} = e^{\mathbf{A}t} \mathbf{C} + e^{\mathbf{A}t} \int_{t_0}^t e^{-\mathbf{A}s} \mathbf{F}(s) ds$$

is a solution of the nonhomogeneous system $\mathbf{X}' = \mathbf{A}\mathbf{X} + \mathbf{F}(t)$,

- $\mathbf{A}$ is an $n \times n$ matrix of constant coefficients,
- $\mathbf{X}$ is an $n \times 1$ matrix (the solution vector),
- $\mathbf{C}$ is an $n \times 1$ matrix (with arbitrary constant entries),

Remark

$$e^{\mathbf{A}t} = \mathcal{L}^{-1} \{ (s\mathbf{I} - \mathbf{A})^{-1} \}$$

$\mathcal{L}^{-1}$ denotes the inverse Laplace transform

Proof:

$\mathbf{X}(t) = e^{\mathbf{A}t}$ is the solution of initial value problem $\mathbf{X}' = \mathbf{A}\mathbf{X}$, $\mathbf{X}(0) = \mathbf{I}$.

$$s\mathbf{x}(s) - \mathbf{X}(0) = \mathbf{A}\mathbf{x}(s) \longrightarrow (s\mathbf{I} - \mathbf{A})\mathbf{x}(s) = \mathbf{I}$$

$$\longrightarrow \mathbf{x}(s) = (s\mathbf{I} - \mathbf{A})^{-1}$$

$$\xrightarrow{\mathcal{L}^{-1}} \mathbf{X}(t) = \mathcal{L}^{-1} \{ (s\mathbf{I} - \mathbf{A})^{-1} \}$$

$$\longrightarrow e^{\mathbf{A}t} = \mathcal{L}^{-1} \{ (s\mathbf{I} - \mathbf{A})^{-1} \}$$

Example: Find $e^{\mathbf{A}t}$ for the matrix

$$\mathbf{A} = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$$

$$e^{\mathbf{A}t} = \mathcal{L}^{-1} \left\{ (s\mathbf{I} - \mathbf{A})^{-1} \right\}$$

Solution

$$s\mathbf{I} - \mathbf{A} = \begin{pmatrix} s & 0 \\ 0 & s \end{pmatrix} - \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} s-2 & 0 \\ 0 & s-3 \end{pmatrix}$$

$$\longrightarrow (s\mathbf{I} - \mathbf{A})^{-1} = \begin{pmatrix} \frac{1}{s-2} & 0 \\ 0 & \frac{1}{s-3} \end{pmatrix}$$

$$\longrightarrow \mathcal{L}^{-1} \left\{ \begin{pmatrix} \frac{1}{s-2} & 0 \\ 0 & \frac{1}{s-3} \end{pmatrix} \right\} = \begin{pmatrix} e^{2t} & 0 \\ 0 & e^{3t} \end{pmatrix} \longrightarrow e^{\mathbf{A}t} = \begin{pmatrix} e^{2t} & 0 \\ 0 & e^{3t} \end{pmatrix}$$

Example: Find $e^{\mathbf{A}t}$ for the matrix

$$e^{\mathbf{A}t} = \mathcal{L}^{-1} \{ (s\mathbf{I} - \mathbf{A})^{-1} \}$$

$$\mathbf{A} = \begin{pmatrix} 1 & -1 \\ 2 & -2 \end{pmatrix}$$

Solution

$$s\mathbf{I} - \mathbf{A} = \begin{pmatrix} s-1 & 1 \\ -2 & s+2 \end{pmatrix}$$

$$\longrightarrow (s\mathbf{I} - \mathbf{A})^{-1} = \begin{pmatrix} s-1 & 1 \\ -2 & s+2 \end{pmatrix}^{-1} = \begin{pmatrix} \frac{s+2}{s(s+1)} & \frac{-1}{s(s+1)} \\ \frac{2}{s(s+1)} & \frac{s-1}{s(s+1)} \end{pmatrix} = \begin{pmatrix} \frac{2}{s} - \frac{1}{s+1} & -\frac{1}{s} + \frac{1}{s+1} \\ \frac{2}{s} - \frac{2}{s+1} & -\frac{1}{s} + \frac{2}{s+1} \end{pmatrix}$$

$$\xrightarrow{\mathcal{L}^{-1}} e^{\mathbf{A}t} = \begin{pmatrix} 2 - e^{-t} & -1 + e^{-t} \\ 2 - 2e^{-t} & -1 + 2e^{-t} \end{pmatrix}.$$

Solution of the Systems of Linear First-Order ODEs

Homogenous

$$\mathbf{X}' = \mathbf{A}\mathbf{X},$$

$$\mathbf{X}(t) = \Phi(t)\mathbf{C}$$

$$\mathbf{X} = e^{\mathbf{A}t}\mathbf{C}$$

Nonhomogenous

$$\mathbf{X}' = \mathbf{A}\mathbf{X} + \mathbf{F}(t),$$

$$\mathbf{X} = \Phi(t)\mathbf{C} + \Phi(t) \int \Phi^{-1}(t)\mathbf{F}(t) dt$$

$$\mathbf{X} = e^{\mathbf{A}t}\mathbf{C} + e^{\mathbf{A}t} \int_{t_0}^t e^{-\mathbf{A}s}\mathbf{F}(s) ds$$

$$e^{\mathbf{A}t} = \mathcal{L}^{-1} \{ (s\mathbf{I} - \mathbf{A})^{-1} \}$$

Exercises 8.3

In problem 1 and 2 find $e^{\mathbf{A}t}$ and $e^{-\mathbf{A}t}$

$$1. \mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$

$$2. \mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

In problem 3 and 4 find $e^{\mathbf{A}t}$

$$3. \mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ -2 & -2 & -2 \end{pmatrix}$$

$$4. \mathbf{A} = \begin{pmatrix} 0 & 0 & 0 \\ 3 & 0 & 0 \\ 5 & 1 & 0 \end{pmatrix}$$

Solution 2. find e^{At} and e^{-At}

$$\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \longrightarrow s\mathbf{I} - \mathbf{A} = \begin{pmatrix} s & -1 \\ -1 & s \end{pmatrix} \longrightarrow (s\mathbf{I} - \mathbf{A})^{-1} = \begin{pmatrix} \frac{s}{s^2 - 1} & \frac{1}{s^2 - 1} \\ \frac{1}{s^2 - 1} & \frac{s}{s^2 - 1} \end{pmatrix}$$
$$= \frac{1}{2} \left(\frac{1}{2} \left\{ \frac{1}{s-1} + \frac{1}{s+1} \right\} \quad \frac{1}{2} \left\{ \frac{1}{s-1} - \frac{1}{s+1} \right\} \right. \\ \left. \frac{1}{2} \left\{ \frac{1}{s-1} - \frac{1}{s+1} \right\} \quad \frac{1}{2} \left\{ \frac{1}{s-1} + \frac{1}{s+1} \right\} \right)$$

$$\longrightarrow e^{At} = \mathcal{L}^{-1} \{ (s\mathbf{I} - \mathbf{A})^{-1} \} = \frac{1}{2} \begin{pmatrix} e^t + e^{-t} & e^t - e^{-t} \\ e^t - e^{-t} & e^t + e^{-t} \end{pmatrix}$$

$$\longrightarrow e^{-At} = \frac{1}{2} \begin{pmatrix} e^{-t} + e^t & e^{-t} - e^t \\ e^{-t} - e^t & e^{-t} + e^t \end{pmatrix} = \frac{1}{2} \begin{pmatrix} e^t + e^{-t} & -(e^t - e^{-t}) \\ -(e^t - e^{-t}) & e^t + e^{-t} \end{pmatrix}$$

In problem 5-10 find the general solution of the given system.

$$5. \mathbf{X}' = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \mathbf{X}$$

$$6. \mathbf{X}' = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \mathbf{X}$$

$$7. \mathbf{X}' = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ -2 & -2 & -2 \end{pmatrix} \mathbf{X}$$

$$8. \mathbf{X}' = \begin{pmatrix} 0 & 0 & 0 \\ 3 & 0 & 0 \\ 5 & 1 & 0 \end{pmatrix} \mathbf{X}$$

$$11. \mathbf{X}' = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$9. \mathbf{X}' = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 3 \\ -1 \end{pmatrix}$$

$$12. \mathbf{X}' = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \mathbf{X} + \begin{pmatrix} \cosh t \\ \sinh t \end{pmatrix}$$

$$10. \mathbf{X}' = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \mathbf{X} + \begin{pmatrix} t \\ e^{4t} \end{pmatrix}$$

Solution 5. first find $e^{\mathbf{A}t}$

$$\mathbf{X}' = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \mathbf{X}$$

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \xrightarrow{\text{diagonal matrix}} e^{\mathbf{A}t} = \begin{pmatrix} e^t & 0 \\ 0 & e^{2t} \end{pmatrix}$$

$$\boxed{\mathbf{X} = e^{\mathbf{A}t} \mathbf{C}} \longrightarrow \mathbf{X}(t) = \begin{pmatrix} e^t & 0 \\ 0 & e^{2t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} c_1 e^t \\ c_2 e^{2t} \end{pmatrix}$$

is the general solution of system

$$\mathbf{X}' = \mathbf{A}\mathbf{X},$$

Solution 7. First find $e^{\mathbf{A}t}$

$$\mathbf{X}' = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ -2 & -2 & -2 \end{pmatrix} \mathbf{X}$$

$$s\mathbf{I} - \mathbf{A} = \begin{pmatrix} s-1 & -1 & -1 \\ -1 & s-1 & -1 \\ 2 & 2 & s+2 \end{pmatrix}$$

$$(s\mathbf{I} - \mathbf{A})^{-1} = \frac{1}{s^3} \begin{pmatrix} s^2 + s & s & s \\ s & s^2 + s & s \\ -2s & -2s & s^2 - 2s \end{pmatrix} = \begin{pmatrix} \frac{1}{s} + \frac{1}{s^2} & \frac{1}{s^2} & \frac{1}{s^2} \\ \frac{1}{s^2} & \frac{1}{s} + \frac{1}{s^2} & \frac{1}{s^2} \\ -\frac{2}{s^2} & -\frac{2}{s^2} & \frac{1}{s} - \frac{2}{s^2} \end{pmatrix}$$

$$\xrightarrow{\mathcal{L}^{-1}} e^{\mathbf{A}t} = \begin{pmatrix} 1+t & t & t \\ t & 1+t & t \\ -2t & -2t & 1-2t \end{pmatrix}$$

$\mathbf{X} = e^{\mathbf{A}t} \mathbf{C}$

is the general solution of system

$$\mathbf{X}' = \mathbf{A}\mathbf{X}, \quad \longrightarrow \quad \mathbf{X}(t) = e^{\mathbf{A}t} \mathbf{c} = \begin{pmatrix} 1+t & t & t \\ t & 1+t & t \\ -2t & -2t & 1-2t \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$$

In problem 13-14 solve the system subject to the given condition.

$$\mathbf{13.} \quad \mathbf{X}' = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ -2 & -2 & -2 \end{pmatrix} \mathbf{X} \quad \text{subject to} \quad \mathbf{X}(0) = \begin{pmatrix} 1 \\ -4 \\ 6 \end{pmatrix}$$

$$\mathbf{14.} \quad \mathbf{X}' = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \mathbf{X} + \begin{pmatrix} 3 \\ -1 \end{pmatrix} \quad \text{subject to} \quad \mathbf{X}(0) = \begin{pmatrix} 4 \\ 3 \end{pmatrix}$$

Solution 13. we already have the general solution

$$\mathbf{X}' = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ -2 & -2 & -2 \end{pmatrix} \mathbf{X}$$

$$\mathbf{X}(0) = \begin{pmatrix} 1 \\ -4 \\ 6 \end{pmatrix}$$

$$\mathbf{X}(t) = e^{\mathbf{A}t} \mathbf{c} = \begin{pmatrix} 1+t & t & t \\ t & 1+t & t \\ -2t & -2t & 1-2t \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$$

$$\mathbf{X}(0) = \mathbf{I} \mathbf{c} = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 1 \\ -4 \\ 6 \end{pmatrix} \longrightarrow c_1 = 1, c_2 = -4, c_3 = 6$$

$$\longrightarrow \mathbf{X}(t) = \begin{pmatrix} 1+t & t & t \\ t & 1+t & t \\ -2t & -2t & 1-2t \end{pmatrix} \begin{pmatrix} 1 \\ -4 \\ 6 \end{pmatrix} = \begin{pmatrix} 1+3t \\ -4+3t \\ 6-6t \end{pmatrix}$$

In problem, 15-18, find $e^{\mathbf{A}t}$ for the coefficient matrix by Laplace transform, then find the general solution of the given system by

$$\mathbf{X} = e^{\mathbf{A}t} \mathbf{C}$$

$$15. \mathbf{X}' = \begin{pmatrix} 4 & 3 \\ -4 & -4 \end{pmatrix} \mathbf{X}$$

$$16. \mathbf{X}' = \begin{pmatrix} 4 & -2 \\ 1 & 1 \end{pmatrix} \mathbf{X}$$

$$17. \mathbf{X}' = \begin{pmatrix} 5 & -9 \\ 1 & -1 \end{pmatrix} \mathbf{X}$$

$$18. \mathbf{X}' = \begin{pmatrix} 0 & 1 \\ -2 & -2 \end{pmatrix} \mathbf{X}$$

Solution 15. First find e^{At}

$$\mathbf{X}' = \begin{pmatrix} 4 & 3 \\ -4 & -4 \end{pmatrix} \mathbf{X}$$

$$s\mathbf{I} - \mathbf{A} = \begin{pmatrix} s-4 & -3 \\ 4 & s+4 \end{pmatrix}$$

$$(s\mathbf{I} - \mathbf{A})^{-1} = \frac{1}{s^2-4} \begin{pmatrix} s+4 & 3 \\ -4 & s-4 \end{pmatrix} = \begin{pmatrix} \frac{3}{4(s-2)} + \frac{1}{4(s+2)} & \frac{3}{4(s-2)} - \frac{3}{4(s+2)} \\ \frac{-1}{s-2} + \frac{1}{s+2} & \frac{-1}{4(s-2)} + \frac{5}{4(s+2)} \end{pmatrix}$$

$$\longrightarrow e^{At} = \begin{pmatrix} \frac{3}{4}e^{2t} + \frac{1}{4}e^{-2t} & \frac{3}{4}e^{2t} - \frac{3}{4}e^{-2t} \\ -e^{2t} + e^{-2t} & -\frac{1}{4}e^{2t} + \frac{5}{4}e^{-2t} \end{pmatrix}$$

$$\longrightarrow \mathbf{X}(t) = e^{At} \mathbf{c} = \begin{pmatrix} \frac{3}{4}e^{2t} + \frac{1}{4}e^{-2t} & \frac{3}{4}e^{2t} - \frac{3}{4}e^{-2t} \\ -e^{2t} + e^{-2t} & -\frac{1}{4}e^{2t} + \frac{5}{4}e^{-2t} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$$